SAGORIKA GHOSH

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Education

University of Washington (UW) Seattle, WA
Master of Science in Data Science, GPA: 3.9/4.0 *Sep 2024 – Mar 2026*

- **Coursework**: NLP, LLM Serving Systems, Deep Learning, Applied Statistics, ML, Data Visualization, Agentic AI, RAG, AWS

Guru Gobind Singh Indraprastha University (GGSIPU) Delhi, India
Bachelor of Technology in Computer Science & Engineering (University Gold Medalist), GPA: 4.0/4.0 *Aug 2018 – Jun 2022*

- **Coursework**: Applied Mathematics, Programming Fundamentals, Probability and Statistics, Machine Learning, DBMS

Professional Experiences

AWS, AI Practitioner *Seattle ; Feb 2026 – Apr 2026*

- Built AI workflows using Amazon Bedrock and SageMaker, including RAG patterns, prompt design, foundation model selection, and inference tuning with latency-cost tradeoff analysis, as part of a selective applied cohort.

Contextual AI, Applied Science Capstone Apprenticeship *Seattle ; Sep 2025 – Mar 2026*

- Built a multilingual multimodal RAG system for HVAC refrigerant recovery, combining WhatsApp voice/image queries, OCR-driven equipment parsing, and grounded QA over 150+ technical manuals.
- Designed the retrieval and serving stack with layout-aware ingestion, metadata/brand-aware filtering, hybrid dense+sparse search, query expansion, and cross-encoder reranking with deduplication, achieving 86.5% Recall@3 and 82.5% MRR@10.
- Benchmarked open-source and Contextual AI pipelines on retrieval, latency, and grounding, using LLM-as-a-Judge for hallucination detection and achieving 2.55s p95 latency in the open-source deployment.

Uber Technologies, Applied Scientist Intern *NYC ; Jun 2025 – Sep 2025*

- Built and deployed Deep Set models (PyTorch, SQL pipelines) for ATD (Actual Time to Delivery) prediction, achieving R^2=0.97, MAE 0.41 (60-min horizon), improving ETA accuracy by 6.2% and reducing courier wait time by 1.7% over XGBoost baselines.
- Used robust statistical methods (IQR outlier filtering, winsorized P50/P75/P90 percentile and stratified KPI analysis across supply regimes) to assess dispatch hold-off decisions under food readiness, ETA, and supply constraints.
- Applied CUPED-based A/B testing to ETA capping policies, reducing UMpT (utilized courier minutes per trip) by 2-3 minutes and improving ETA prediction accuracy by 2.1% in undersupplied markets.

American Express, Data Scientist 2 *Aug 2023 – Aug 2024*

- Improved fraud labeling (Suspect Claims, Recurring Charges) by optimizing XGBoost models, increasing recall by 1.2% at the top 0.2% risk band and driving about $20K in annual fraud loss reduction.
- Developed an LSTM model to predict peak resource demand with 98.9% accuracy to support cloud autoscaling during high-traffic events (eg: Black Friday), reducing pod ramp-up time by 53.2% compared to manual ramp-up.
- Built and deployed LSTM-based forecasting models (CNN-LSTM for FX rates; BiLSTM for inbound call volumes across 20 categories), improving prediction accuracy by up to 31.4% and supporting secure deployment across 28 global markets.

American Express, Data Engineer *Aug 2022 – Aug 2023*

- Migrated 75% of GAM data to AmEx's big data lake, optimizing ETL pipelines (12% faster processing, 23% lower latency) and improving troubleshooting by 48%.
- Built Java microservices to migrate 82% IMS transactions to cloud by routing ∼200K GAM requests/day, reducing infrastructure costs by $10.2K annually and enabling scalable credit fraud and risk analytics.

Hitachi Vantara, ML Engineer *Feb 2022 – Jul 2022*

- Optimized IQ data pipelines by integrating U-Net autoencoder-based anomaly detection, improving data processing by 12.3%.

Microsoft, Engage Intern *Oct 2021 – Dec 2021*

- Built a neural collaborative filtering scheduling recommender with constraint optimization, improving scheduling efficiency by 29%.

Technical Skills

ML & Modeling: Regression, Classification, Time-Series Forecasting, NLP, Deep Learning, Transformers
LLMs & GenAI: RAG, Agentic AI, CrewAI (multi-agent systems), Prompting, Evaluation
Systems & Data: Python, C++, Java, SQL, NoSQL (MongoDB), Spark, Hive, ETL, AWS (Bedrock, SageMaker), Databricks, Snowflake
Experimentation: A/B Testing, CUPED, Causal Inference
Tools: Git, CI/CD, Jenkins, Tableau, Streamlit, Agile Methodology

Academic Projects

RAG-OS (Open-Source RAG Pipeline)Demo *Feb 2026*

- Built a production-grade RAG system for technical HVAC manuals using Docling, Qdrant, and BGE embeddings; implemented adaptive chunking and cross-encoder reranking to achieve 75% Recall@5 in procedure-level retrieval.
- Designed a rigorous evaluation framework (LLM-as-Judge) to measure Faithfulness, Relevance, and Completeness, reaching a 4.38/5 quality score while reducing evaluation time by 75% through parallelization and resilient API handling.

US Census Chat Agent *Feb 2026*

- Created an agentic text-to-SQL Census assistant using Snowflake Cortex with cryptographic-style SQL validation, semantic vector guardrails & a deterministic fallback router, achieving 95.6% heuristic query execution success on complex demographic queries.

Quant Copilot *Nov 2025*

- Architected a RAG-based investment analysis system on Databricks, serving Llama inference over Delta tables (5M+ rows) with 1.8s median latency and demonstrating a 60% end-to-end time reduction in controlled manual vs. RAG-assisted benchmarks.